Read Me

The following Git Hub folder contains 4 folders and 2 documents.

Folders:

1. Fnc_stats
2. gica_cmd_mancovan_results_summary
3. Sm_stats
4. Spectra_stats

Documents:

1. gica_cmd_mancovan.mat
2. gica_cmd_mancovan_mancovan_results.log

All the Independent Components utilized in the following folders use the Group Independent Component Analyses (GICA) Nomenclature, which corresponds to the Neuromark Nomenclature we used throughout the manuscript as follows:

GICA	Neuromark
1	69
3	98
4	99
5	45
6	21
8	3
9	9
10	2
11	11
13	54
14	66
16	72
22	93
26	68
27	33
28	43
29	70
31	55
34	84
36	88
40	67

43	32
48	51
50	13
51	18
52	4
53	7

In the following documents, the label "AttentionNetwork" is a placeholder automatically assigned by the GICA pipeline and does not reflect the true functional role of the network. The abbreviation PAT refers to patients with Parkinson's disease (PwPD) as described in the manuscript, while CTRL refers to healthy older adult controls (HOA).

Abbreviations:

PAT: patients

CTRL: controls

FNC: functional network connectivity

ICs: Independent Components

The following section describes the information found within the following folders and documents:

1. **Fnc_stats**

gica_cmd_mancovan_results_fnc.mat: includes the connectivity matrices derived from GICA time courses and the results (t-tests, F-tests, covariate effects) of group-level statistical comparisons.

gica_cmd_mancovan_results_summary

icatb_mancovan_results_summary_01.png.-icatb_mancovan_results_summary_26.png AND icatb_mancovan_results_summary_28.png- icatb_mancovan_results_summary_55.png:

The figures include spatial maps and corresponding power spectra for the selected ICs. Each spatial map highlights brain regions with the strongest activity, with peak activation shown in yellow and surrounding clusters at lower intensity in red. Below each map, the associated power spectrum illustrates the frequency profile of the component time series, typically characterized by a dominant peak at low frequencies and a gradual decline at higher frequencies. Together, the spatial maps and spectra confirm both the anatomical localization and the expected frequency dynamics of each IC and confirm the reliability of the functional connectivity results.

icatb_mancovan_results_summary_27.png: FNC Correlation Matrix Averaged Across Subjects. The heatmap displays pairwise correlation values among all selected GICA ICs. Warmer colors (red/yellow): stronger positive correlations; cooler colors (blue): stronger negative correlations. Values are averaged across all participants, providing an overview of group-level functional connectivity patterns among brain networks.

icatb_mancovan_results_summary_56.png: The image shows statistical group differences between PAT and CTRL. Voxels in red-yellow indicate regions with significant group differences in connectivity strength (displayed as −log10 p-values).

icatb_mancovan_results_summary_57.png: This figure shows spectral F-test results comparing patients PAT and CTRL. The x-axis represents frequency (Hz), the y-axis lists ICs, and the color scale indicates –log10(p-value). Warm colors (red/yellow) highlight significant group differences, here displayed at a threshold of $p < 0.01$, primarily in the low-frequency range (<0.1 Hz).

icatb_mancovan_results_summary_58.png: The matrix shows significant FNC differences at $p < 0.01$. Each square represents a pair of ICs, with color indicating the direction and magnitude of group differences (scale = $-\mathrm{sign}(t)\log_{10}(p)$).

icatb_mancovan_results_summary_59.png: Connectogram illustrating significant (uncorrected $p<0.01$) correlations between ICs. Yellow lines indicate significant FNC differences between PAT and CTRL groups, with color scale reflecting $-\log_{10}(p)$ where there is an increase in functional connectivity in PwPD versus controls. Red lines between the brain regions indicate that CTRL had higher functional connectivity than PAT. Each IC is represented as an area surrounding the connectogram.

icatb_mancovan_results_summary_60.png: This figure presents significant spatial and spectral effects of the comparison between PAT and CTRL groups within the network of interest. The top row shows spatial clusters with significant group differences ($p < 0.01$, thresholded on a $-\log_{10}(p)$ scale). The bottom panel displays effect size estimates (average beta values) for each IC, with bar lengths representing the magnitude of the effect and colors reflecting the fraction of the component affected. Positive beta values indicate greater connectivity or activation in PAT relative to CTRL, while negative beta values indicate reduced connectivity or activation in PAT relative to CTRL.

icatb_mancovan_results_summary_61.png: This figure summarizes significant group differences between PAT and CTRL. The top panel shows frequency spectra differences across components, with color scale reflecting significance levels ($-\log_{10}$ p-value). Warm colors (red–yellow) indicate frequencies where PAT> CTRL, and cool colors (blue) indicate the opposite. The bottom panel shows effect sizes (average beta weights) across components, with horizontal bars representing both direction and magnitude of group differences, normalized by the fraction of spectrum variance explained.

icatb_mancovan_results_summary_62.png: This figure displays significant group differences in FNC between PAT and CTRL for the Attention Network ($p < 0.01$). The heatmap shows pairwise component comparisons, with the color scale indicating the significance level (−log10 p-value). Warm colors (red/yellow) denote stronger group effects, while cool colors (blue) indicate negative or opposite effects. Each axis lists the ICs compared, and only connections surviving the $p < 0.01$ threshold are shown.

icatb_mancovan_results_summary_63.png: This connectogram illustrates significant connectivity differences between PAT and CTRL. Each IC is shown around the circle with its corresponding spatial map. Curved lines indicate connections between components where group differences reached statistical significance ($p < 0.01$). The color of each line represents the strength and direction of the effect, coded by –sign(t)·log10(p): warm colors (yellow–red) denote stronger or increased connectivity, while cool colors (blue) denote reduced connectivity. The bottom color bar provides the scale.

2. **Sm_stats**

Contains spatial map–based statistical results from the Mancovan analysis.

Beta weight files (.mat) provide regression coefficients quantifying group or covariate effects on IC spatial maps.

Significant voxel wise effects are stored as statistical maps in Analyze format (.hdr/*.img pairs), highlighting brain regions where IC spatial maps differ significantly between groups or conditions.

3. **Spectra_stats**

This folder contains group comparison results for the frequency-domain properties of IC time courses.

...*betas**.mat: Beta weights (regression coefficients) indicating group differences in spectral power across ICs.

...*results_spectra**.mat: Statistical outputs (F-values, p-values, effect sizes) summarizing group differences for each IC and frequency band tested.

Documents:

1. **gica_cmd_mancovan.mat**

This MATLAB file stores the main results and configuration data from the GIFT Multivariate Analysis of Covariance (MANCOVAN) run. It contains the statistical outcomes, model parameters, and links to other result files (e.g., sm_stats, fnc_stats, spectra_stats). Use it as the entry point for re-loading a full MANCOVAN analysis into the GIFT GUI or MATLAB workspace.

2. **gica_cmd_mancovan_mancovan_results.log**

This log file documents the execution of the GICA Mancovan analysis, including subject loading, time course preprocessing (despiking, filtering), computation of spatial maps, spectra, and FNC, along with the corresponding t-statistics and p-values. It provides a chronological record of which ICs were analyzed, the steps performed, and the statistical outputs saved to the sm_stats, spectra_stats, and fnc_stats folders